Safe steering or near-identity reranking?

A case study of text detox with embedding-guided candidate selection

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Research note prepared from the week 12 text-detox reranking experiment

Take-home message: sentence embeddings contain a usable detox signal, but automatic reranking collapses toward near-identity outputs; the main bottleneck is candidate generation and conservative selection.

Code & Paper

1. MOTIVATION & QUESTION

Text detoxification should reduce toxicity while preserving meaning. We test whether toxicity-related directions in pretrained sentence embeddings can guide candidate selection, and whether local directions help more than a single global direction.

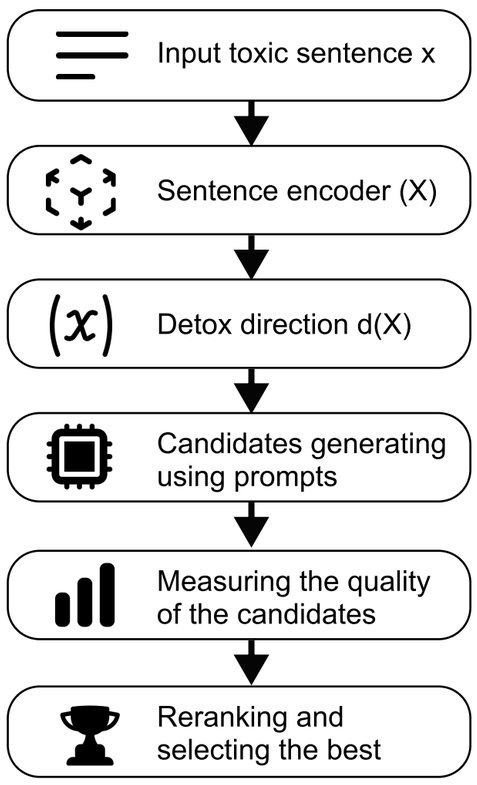
Example

"give me a fucking list on one page assholes"	→	"get me plenty lists, please that would take too mesquitle up"
Toxic message		**Detox message**

2. METHOD

Pipeline algorithm

1. Encode the input text using a sentence encoder E(x).
2. Construct an embedding-space target: $z' = E(x) + \alpha d(x)$ where d(x) is either a global detox direction or a cluster-local one.
3. Generate a bank of candidate rewrites with a prompt-based text rewriter.
4. Score each candidate using target similarity, toxicity reduction, semantic similarity, and a surface penalty.
5. Select the best candidate using either fixed α = 1.0 or automatic α*.



Important note that we don't decode directly steered text vector

Automatic a* selection function

$$J(c;\alpha) = w_t \cdot target_sim + w_d \cdot \Delta tox + w_s \cdot sim - w_p \cdot surf$$

3. EXPERIMENTAL SETUP

TRAINING DATA
- Filtered Civil Comments subset
- 9,942 training rows for direction learning

EVALUATION DATA
- 300 toxic Civil Comments examples
- 150 ParaDetox examples

MODELS
- Sentence encoder: all-mpnet-base-v2
- Toxicity scorer: unitary/toxic-bert
- Rewriter: google/flan-t5-base

LOCAL MODELING
- MiniBatch K-Means with K = 8
- Separate local detox direction per cluster

SELECTION SETTINGS
- Alpha grid: {0, 0.5, 1.0, 1.5, 2.0}
- Fixed policy: α = 1.0
- Automatic policy: best feasible α*

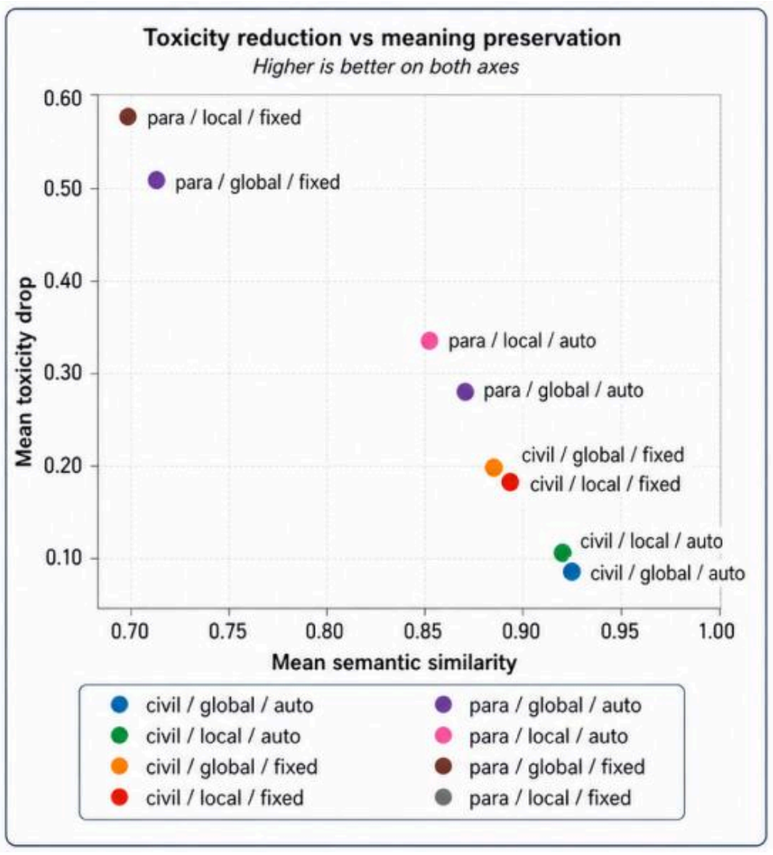
FEASIBILITY CONSTRAINTS
- Semantic similarity ≥ 0.72
- Toxicity drop ≥ 0.12

METRICS
- Mean toxicity drop
- Mean semantic similarity
- Acceptance rate
- Mean selected alpha

PLATFORMS USED

Kaggle | Jupyter Notebook | Python

4. MAIN RESULTS



Each point is one reranking setting. The x-axis shows how much original meaning is preserved; the y-axis shows how much toxicity is reduced. The ideal method should be in the upper-right corner: high detoxification and high meaning preservation.

- Fixed scoring with α = 1.0 produces stronger detoxification, especially on ParaDetox.
- The best toxicity reduction is obtained by Local Fixed on ParaDetox: toxicity drop = 0.5850.
- However, this comes with lower semantic similarity: 0.6960. This means the model changes the text more aggressively.

The pipeline can reduce toxicity, but the best detoxification results are obtained by more aggressive fixed scoring, not by the automatic policy.

The comparison table (higher is better for both)

Civil Comments		
Setting	**Tox drop**	**Sem sim**
Global Auto	0.0927	0.9563
Global Fixed	0.1923	0.8912
Local Auto	0.1093	0.9498
Local Fixed	0.1836	0.9004

ParaDetox		
Setting	**Tox drop**	**Sem sim**
Global Auto	0.2775	0.8731
Global Fixed	0.5139	0.7244
Local Auto	0.3332	0.8512
Local Fixed	0.5850	0.6960

- Toxicity is the score from Toxic-BERT; higher means stronger detoxification.

Tox drop = toxicity(original) − toxicity(rewritten)

- Sem sim measures how close the rewritten sentence remains to the original meaning.

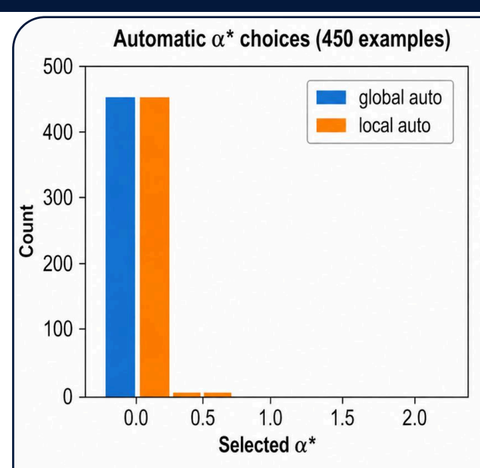
$$\text{Sem sim} = \cos(E(x), E(c))$$

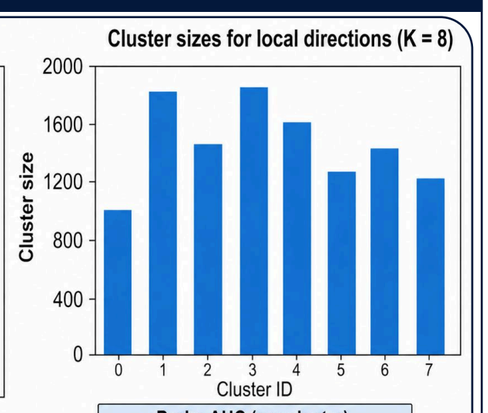
The results show a clear trade-off: stronger detoxification usually reduces meaning preservation.

5. LOCAL VS GLOBAL

The automatic policy was expected to choose a safe editing strength for each sentence. Instead, it almost always selects α* = 0.

Global Auto chooses zero in 100% of cases; Local Auto chooses zero in 98.89% of cases. This means the method often prefers no edit or a very weak edit.





Probe AUC (per cluster)	
Min	0.8745
Max	0.9482
Fallback rate	0.000

Local directions sometimes improve the score, especially on ParaDetox, but the improvement is smaller than the gap between Auto and Fixed scoring. Therefore, the main issue is not simply global vs local steering, it is that the selector is too conservative and often keeps the original sentence.

Local steering is technically healthy, but automatic reranking collapses toward near-identity outputs.

6. CONCLUSIONS & NEXT STEPS

Key conclusions

1. **Embeddings contain a detox signal.** A simple toxicity probe reaches AUC = 0.8931, showing that toxic and clean texts are separable in sentence-embedding space.
2. **Fixed scoring works, but it is aggressive.** Fixed α = 1.0 gives larger toxicity reduction, especially on ParaDetox, but also lowers semantic similarity.
3. **Automatic selection is too conservative.** Auto α* usually selects 0, so the system often returns the original text or a near-identity rewrite.
4. **Local directions are stable, but not decisive.** All 8 local clusters are usable, yet local steering does not solve the main bottleneck: weak candidate generation and conservative selection.

Next steps

1. **Penalize no-op outputs.** Do not allow the original sentence to win unless it achieves a real toxicity reduction.
2. **Improve candidate generation.** Use a stronger rewriter or better prompts so that the candidate bank contains more useful detoxified rewrites.
3. **Report nontrivial edit rate.** Measure how often the selected output is meaningfully different from the input, not only whether the score improves.
4. **Align evaluation with the toxicity scorer.** Build test subsets where examples are toxic according to the same scorer used for evaluation.
5. **Run multi-seed experiments.** Repeat the comparison across several random seeds before making final claims about global vs local steering.